

Lema de DuBois Reymond

Lema 1 (DuBois Reymond). Sea $M \in \mathcal{C}([a, b])$ tal que

$$\forall \varphi \in \mathcal{C}^1([a, b]) : \varphi(a) = 0 = \varphi(b) : \int_a^b M(x)\varphi(x) \, dx = 0 \implies M \equiv 0 \text{ en } [a, b].$$

Demostración: Suponemos por contradicción que $\exists x_0 \in (a, b) : M(x_0) \neq 0$ (sin pérdida de generalidad, suponemos $M(x_0) > 0$). Entonces, como M es continua, sabemos que existen x_1, x_2 con $a < x_1 < x_0 < x_2 < b$ tales que $M(x) > 0$ para $x \in [x_1, x_2]$.

$$\text{Tomamos } \varphi(x) := \begin{cases} (x - x_1)^2(x_2 - x)^2 & \text{si } x \in [x_1, x_2], \\ 0 & \text{resto.} \end{cases}$$

$$\implies 0 = \int_a^b M(x)\varphi(x) \, dx = \int_{x_1}^{x_2} M(x)\varphi(x) \, dx > 0. \quad \longrightarrow \longleftarrow$$

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Referenciado en

- teo-euler-lagrange

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